

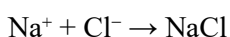
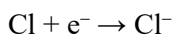
## Bonding

### 1. What is chemical bonding? Explain ionic, covalent, and metallic bonds. ★

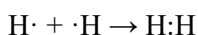
Chemical bonding is the attractive force that holds atoms or ions together in a compound. Atoms form bonds to achieve a more stable electron arrangement, usually a complete outer shell or an octet of 8 valence electrons.

Atoms may lose, gain, or share electrons.

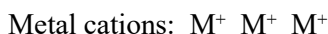
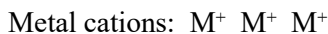
An ionic bond is the electrostatic attraction between oppositely charged ions. It usually forms between a metal and a non-metal. The metal loses electron(s) and forms a cation; the non-metal gains electron(s) and forms an anion.



A covalent bond is formed when two atoms share one or more pairs of electrons. It usually forms between two non-metals.



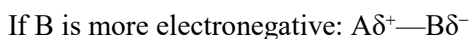
A metallic bond is the attraction between positive metal ions and a mobile sea of delocalized electrons.



### 2. How is a chemical bond formed? What is the role of electronegativity? ★

A chemical bond forms when atoms interact through their valence electrons to become more stable. Electrons may be transferred, shared, or delocalized.

Electronegativity is the ability of an atom to attract shared electrons. Metals generally have low electronegativity, while non-metals generally have high electronegativity.



Nearly zero electronegativity difference gives a non-polar covalent bond; a moderate difference gives a polar covalent bond; and a very large difference gives an ionic bond.

### 3. What are single and double covalent bonds? ★

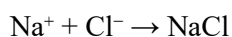
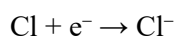
A single bond is formed by sharing one pair of electrons. It contains one sigma ( $\sigma$ ) bond and is shown by one line:  $\text{H}—\text{H}$ .

A double bond is formed by sharing two pairs of electrons. It contains one sigma ( $\sigma$ ) bond and one pi ( $\pi$ ) bond and is shown by two lines:  $\text{O}=\text{O}$ .

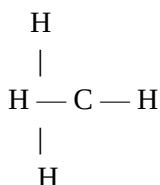
| Feature               | Single bond | Double bond          |
|-----------------------|-------------|----------------------|
| Shared electron pairs | 1           | 2                    |
| Bonds present         | 1 $\sigma$  | 1 $\sigma$ + 1 $\pi$ |
| Strength              | Weaker      | Stronger             |
| Bond length           | Longer      | Shorter              |

#### 4. What type of bond is present in NaCl and methane?

NaCl contains an ionic bond.



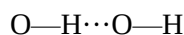
Methane ( $\text{CH}_4$ ) contains covalent C—H bonds. These bonds are slightly polar covalent, but methane is overall non-polar because of its symmetrical shape.



#### 5. If hydrogen bonding is present, how are boiling point and melting point affected? ★

Hydrogen bonding is a strong intermolecular attraction that occurs when hydrogen is directly bonded to fluorine, oxygen, or nitrogen.

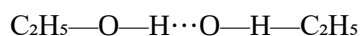
It generally increases boiling point and melting point because more energy is needed to separate the molecules.



The dotted line represents a hydrogen bond.

#### 6. What are the differences between hydrogen bonding in ethanol and methane? ★

Ethanol ( $\text{C}_2\text{H}_5\text{OH}$ ) has an O—H group, so its molecules can form hydrogen bonds. It can donate hydrogen bonds through O—H and accept them through oxygen.



Methane ( $\text{CH}_4$ ) has no O—H, N—H, or F—H bond, so it cannot form hydrogen bonds. Its main intermolecular force is London dispersion force.

| Property               | Ethanol         | Methane    |
|------------------------|-----------------|------------|
| Hydrogen bonding       | Present         | Absent     |
| Boiling point          | Higher          | Much lower |
| Water solubility       | Relatively high | Very low   |
| Room-temperature state | Liquid          | Gas        |

#### 7. What is metallic bonding? Explain the properties of metals. ★

A metallic bond is the attraction between positive metal ions and a mobile sea of delocalized electrons.

Metals conduct electricity because free electrons move through the metal. They conduct heat because these electrons transfer kinetic energy. Metals are malleable and ductile because the layers can slide while the metallic bond remains. Metals are lustrous because their delocalized electrons reflect light.

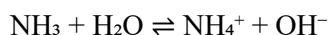
### Acids and Bases

#### 1. Give the definitions of acids and bases. ★

Arrhenius acid: a substance that produces  $\text{H}^+$  in water. CODE:  $\text{HCl} + \text{H}_2\text{O} \rightarrow \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$

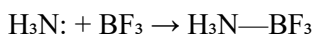
Arrhenius base: a substance that produces  $\text{OH}^-$  in water. CODE:  $\text{NaOH} + \text{H}_2\text{O} \rightarrow \text{Na}^+(\text{aq}) + \text{OH}^-(\text{aq})$

Brønsted–Lowry acid: a proton donor. Brønsted–Lowry base: a proton acceptor.



$\text{NH}_3$  accepts  $\text{H}^+$ , so it is a base;  $\text{H}_2\text{O}$  donates  $\text{H}^+$ , so it is an acid.

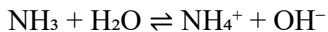
Lewis acid: an electron-pair acceptor. Lewis base: an electron-pair donor.



$\text{NH}_3$  is the Lewis base and  $\text{BF}_3$  is the Lewis acid.

## 2. What is a conjugate acid and a conjugate base? ★

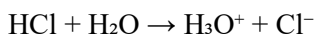
A conjugate acid is formed when a base accepts a proton. A conjugate base is formed when an acid donates a proton.



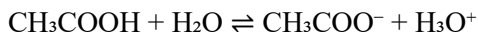
$\text{NH}_4^+$  is the conjugate acid of  $\text{NH}_3$ , and  $\text{OH}^-$  is the conjugate base of  $\text{H}_2\text{O}$ .

## 3. Define strong acid and weak acid. ★

A strong acid is completely ionized in water. Examples are  $\text{HCl}$ ,  $\text{HBr}$ ,  $\text{HI}$ ,  $\text{HNO}_3$ ,  $\text{H}_2\text{SO}_4$ , and  $\text{HClO}_4$ .



A weak acid is only partially ionized in water and establishes equilibrium. Examples are  $\text{CH}_3\text{COOH}$ ,  $\text{HCOOH}$ ,  $\text{HF}$ ,  $\text{HCN}$ ,  $\text{H}_2\text{S}$ , and  $\text{H}_2\text{O}$ .



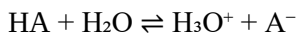
Strength refers to the degree of ionization; concentration refers to the amount of solute present.

## 4. What are equilibrium constant and acid dissociation constant, $K_a$ ? ★

For  $aA + bB \rightleftharpoons cC + dD$ , the equilibrium constant is:

$$K_c = [\text{C}]^c[\text{D}]^d / [\text{A}]^a[\text{B}]^b$$

The acid dissociation constant,  $K_a$ , is the equilibrium constant for ionization of a weak acid in water.



$$K_a = [\text{H}_3\text{O}^+][\text{A}^-] / [\text{HA}]$$

For acetic acid:  $K_a = [\text{H}_3\text{O}^+][\text{CH}_3\text{COO}^-] / [\text{CH}_3\text{COOH}] = 1.75 \times 10^{-5}$

A larger  $K_a$  means a stronger acid; a smaller  $K_a$  means a weaker acid.

## Electrochemistry

### 1. Define electrochemistry, voltage, charge, and current. ★

Electrochemistry is the branch of chemistry that studies the relationship between chemical reactions and electricity. It includes producing electricity from spontaneous reactions and using electricity to drive non-spontaneous reactions.

Voltage is electric potential difference; its SI unit is volt (V). Electric charge causes attraction or repulsion between charged particles; its unit is coulomb (C). Electric current is the flow of charge per unit time.

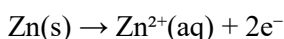
$$I = Q / t$$

$$1 \text{ A} = 1 \text{ C/s}$$

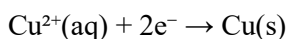
$$\text{Energy (J)} = \text{Potential (V)} \times \text{Charge (C)}$$

### 2. What are the two types of electrodes? Explain their reactions. ★

Anode: oxidation occurs at the anode. Oxidation is loss of electrons.



Cathode: reduction occurs at the cathode. Reduction is gain of electrons.



Important rule: Anode = Oxidation; Cathode = Reduction. Memory aid: OIL RIG (Oxidation Is Loss, Reduction Is Gain).

### 3. What is an electrochemical cell? Explain its types. ★

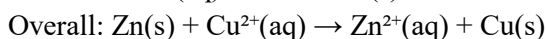
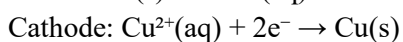
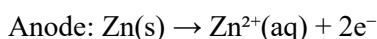
An electrochemical cell is a system in which chemical energy and electrical energy are converted into each other. It generally contains an anode, cathode, electrolyte, external wire, and sometimes a salt bridge.

A galvanic or voltaic cell converts a spontaneous chemical reaction into electrical energy. An electrolytic cell uses external electrical energy to force a non-spontaneous reaction.

| Feature           | Galvanic cell         | Electrolytic cell           |
|-------------------|-----------------------|-----------------------------|
| Reaction          | Spontaneous           | Non-spontaneous             |
| Energy conversion | Chemical → electrical | Electrical → chemical       |
| External power    | Not required          | Required                    |
| Anode sign        | Negative              | Positive                    |
| Cathode sign      | Positive              | Negative                    |
| Example           | Daniell cell          | Electrolysis of molten NaCl |

### 4. Explain anode, cathode, oxidation, and reduction reactions. ★

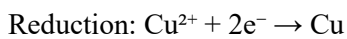
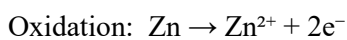
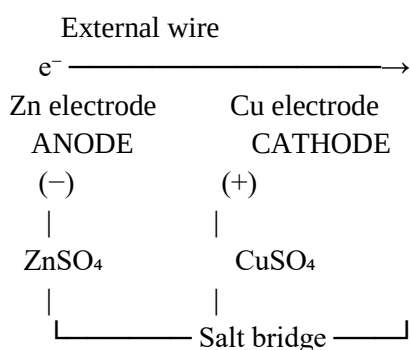
At the anode, zinc atoms lose electrons and enter solution as  $\text{Zn}^{2+}$ . At the cathode, copper ions gain electrons and deposit as solid copper.



This is a redox reaction because oxidation and reduction always occur together.

### 5. Explain an electrochemical cell using the Daniell cell. ★

A Daniell cell contains a zinc electrode in  $\text{ZnSO}_4$  solution, a copper electrode in  $\text{CuSO}_4$  solution, a connecting wire, and a salt bridge.



The salt bridge allows ion movement, maintains electrical neutrality, completes the internal circuit, and prevents rapid mixing. Electrons flow through the external wire from zinc to copper.

### 6. What is the difference between a galvanic cell and an electrolytic cell?

A galvanic cell produces electricity from a spontaneous reaction, whereas an electrolytic cell uses electricity to drive a non-spontaneous reaction. Oxidation still occurs at the anode and reduction still occurs at the cathode in both cells; only the electrode signs change.

| Point        | Galvanic cell        | Electrolytic cell    |
|--------------|----------------------|----------------------|
| Purpose      | Produces electricity | Uses electricity     |
| Anode sign   | Negative             | Positive             |
| Cathode sign | Positive             | Negative             |
| Example      | Daniell cell         | Electrolysis of NaCl |

### 7. What is transport number? Give its mathematical expression. ★

The transport number, or transference number, of an ion is the fraction of the total electric current carried by that ion.

$$t_+ = I_+ / I$$

$$t_- = I_- / I$$

$$I = I_+ + I_-$$

Therefore:

$$t_+ = I_+ / (I_+ + I_-)$$

$$t_- = I_- / (I_+ + I_-)$$

$$t_+ + t_- = 1$$

In terms of ionic conductance:

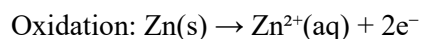
$$t_+ = \lambda_+ / (\lambda_+ + \lambda_-)$$

$$t_- = \lambda_- / (\lambda_+ + \lambda_-)$$

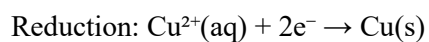
Example: if  $I_+ = 0.40$  A and  $I_- = 0.60$  A, then  $I = 1.00$  A,  $t_+ = 0.40$ , and  $t_- = 0.60$ .

### 8. Calculate the standard cell potential for the Daniell cell. ★

Given:  $E^\circ_{\text{red}}(\text{Cu}^{2+}/\text{Cu}) = +0.339$  V and  $E^\circ_{\text{red}}(\text{Zn}^{2+}/\text{Zn}) = -0.762$  V.



$$E^\circ_{\text{oxidation}} = -(-0.762 \text{ V}) = +0.762 \text{ V}$$



$$E^\circ_{\text{reduction}} = +0.339 \text{ V}$$

$$E^\circ_{\text{cell}} = E^\circ_{\text{reduction}} + E^\circ_{\text{oxidation}}$$

$$E^\circ_{\text{cell}} = 0.339 \text{ V} + 0.762 \text{ V}$$

$$E^\circ_{\text{cell}} = +1.101 \text{ V}$$

Therefore, the standard cell potential is +1.101 V.